QUESTION 1

a) **(i)** Name structures **X** and **Y**. [2]

X: Heterochromatin

Y: Euchromatin

(ii) Account for the difference in the structures **X** and **Y**. [2]

1. Region X is more electron dense / darkly stained than region Y
2. Thus heterochromatin is more condensed / tightly packed / coiled than euchromatin
3. Heterochromatin is transcriptionally inactive as compared to euchromatin
4. Heterochromatin is wound around deacetylated histones

Fig. 1.2 shows part of a DNA molecule.

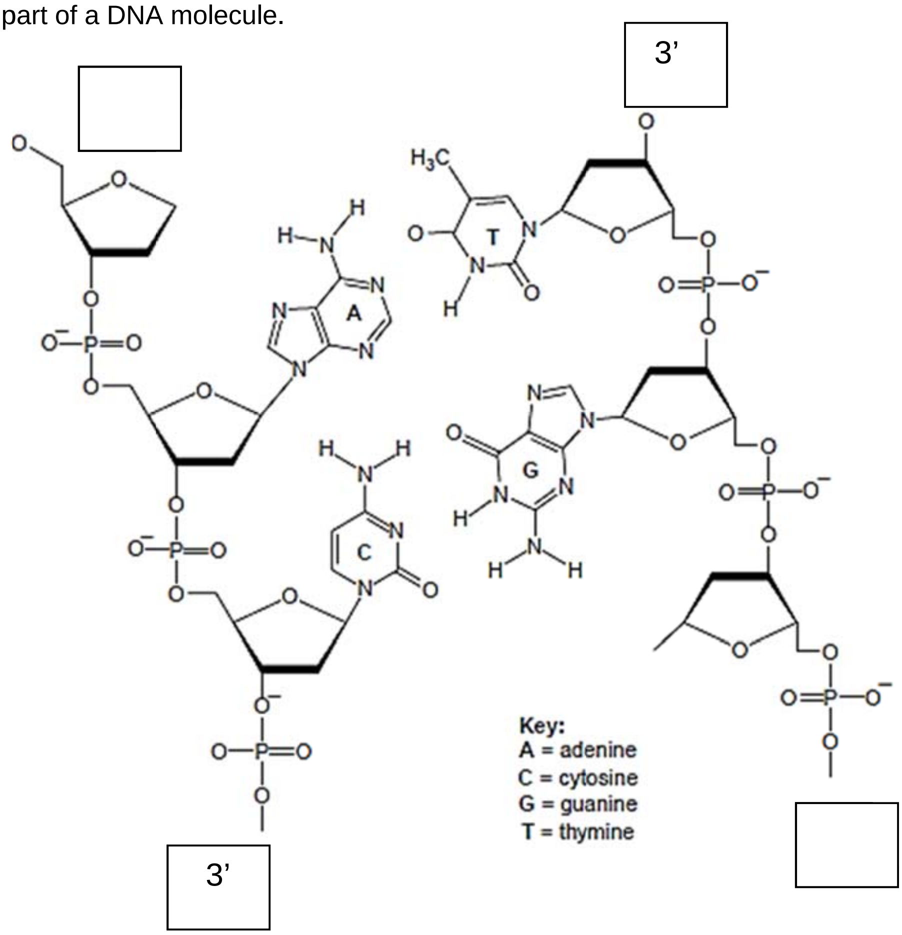

Fig. 1.2

b) **(i)** Complete Fig. 1.2 by indicating the polarity of the DNA molecule in the boxes provided. [1]

(ii) State the importance of hydrogen bonds in DNA structure. [2]

1. Hydrogen bonds allow for the formation of double stranded DNA / a double helix
2. Hydrogen bonds hold the two polynucleotide strands together
3. Hydrogen bonds hold the complementary nucleotides / bases together
4. Many hydrogen bonds give stability to DNA molecule
5. Hydrogen bonds can be broken for transcription / DNA replication to occur

c) Explain how the data in Table 1.1 helps to confirm the arrangement of bases in DNA. [3]

1. In all the organisms, percentage of adenine to thymine and percentage of guanine to cytosine are approximately 1:1 ratio / equal / similar
2. For example, in yeast, percentage of adenine is 31.3% which is very similar to thymine with percentage at 32.9% and percentage of guanine is 18.7% which is very similar to cytosine with percentage at 17.1%
   Accept any one example
3. This shows that adenine and guanine base pair with thymine and cytosine respectively

d) **(i)** State how the result for the virus differs from those for all the organisms given in Table 1.1. [1]

1. The percentage of adenine to thymine and guanine to cytosine are not similar / not 1:1

(ii) Give a reason for your answer to **(d) (i)**. [1]

1. Virus is made up of single-stranded DNA / DNA that is not a double helix

[Total: 12 marks]

QUESTION 2

(a) Describe how an individual can albinism from his parents. [2]

1. Both parents must be carriers / affected / 1 carrier and 1 affected / has at least 1 recessive allele each
2. The individual must receive 1 recessive allele from each parent / 2 recessive alleles, 1 from each parent

(b) Using the symbols provided, draw a genetic diagram in the space below to show the cross between the couple. [5]

Let **N** represent the dominant allele for normal skin pigment
n represent the recessive allele for no skin pigment
B represent the dominant allele for normal red-green colour vision
b represent the recessive allele for red-green colour blindness

Parental phenotype	Man with albino & normal colour vision	x	Woman with normal skin pigment & normal colour vision
Parental genotypes	$nn\,X^BY$	x	NnX^BX^b
Gametes	nX^B, nY		NX^B, NX^b, nX^B, nX^b

Random fertilization

		Female gametes			
		NX^B	NX^b	nX^B	nX^b
Male gametes	nX^B	NnX^BX^B	NnX^BX^b	nnX^BX^B	nnX^BX^b
	nY	NnX^BY	NnX^bY	nnX^BY	nnX^bY

Offspring Genotypic Ratio: 1 NnX^BX^B : 1 NnX^BX^b : 1 nnX^BX^B : 1 nnX^BX^b : 1 NnX^BY : 1 NnX^bY : 1 nnX^BY : 1 nnX^bY

Offspring Phenotypic Ratio: 2 normal skin pigment & normal colour vision female : 2 albino & normal colour vision female : 1 normal skin pigment & normal colour vision male : 1 normal skin pigment & colour blind male : 1 albino & normal colour vision male : 1 albino & colour blind male

[Total: 7 marks]

QUESTION 3

(a) **(i)** Describe the effect of flashing light on ATP synthesis in the absence of venturicidin. [2]

1. As the number of light flashes increases from 1 – 3 flashes, the concentration of ATP increases from 25nM – 190 nM
2. which eventually plateaus

(ii) Explain your answer in **a (i)** [4]

1. Increasing the number of flashes of light result in an increase in the number of excited electrons
2. and results in more excited electrons travelling down the electron transport chain
3. The energy released is coupled to pump/ actively transport H^+
4. from the stroma into the thylakoid space/ thylakoid lumen
5. resulting in the formation of a proton motive force/ electrochemical gradient
6. H^+ then diffuses down the concentration gradient from the thylakoid space to the stroma
7. via the ATP synthase complex
8. releasing more free energy which is coupled for synthesise more ATP

(b) Describe and explain the effect of venturicidin on the plant and its ability to produce sugars. [4]

1. In the presence of ventruricidin, ATP concentration remains at close to 0nM even when 3 flashes of light are used
2. Since ATP synthase is inhibited, H^+ cannot diffuse down the electrochemical gradient
3. from thylakoid space to stroma
4. thus there is no supply of energy to drive ATP synthesis/ no energy released for formation of ATP
5. This prevents the progression of the Calvin's cycle / no G3P/ no hexose sugar being produced
6. since ATP is not available for reduction of GP to 1,3-bisphosphoglycerate / regeneration of RuBP

(c) Distinguish the role of ATP synthesis in mitochondria and chloroplasts [1]

1. In mitochondria, ATP synthesised is transported into the cytosol and used for various cellular functions/ active transport/ protein synthesis/ @ref to specific process
2. While ATP synthesised in chloroplasts is transported into the stroma for use in the light-independent reaction / Calvin cycle for synthesis of glucose

[Total: 11 marks]

QUESTION 4

a) Describe how the gene coding for bovine somatotrophin is inserted into the plasmid. [3]

1. Cleave with the same restriction enzyme
2. To generate sticky-ended fragments flanking the bovine somatotrophin gene
3. And linearise plasmid vector at the restriction site
4. Within tetracycline resistance gene
5. To generate complementary sticky ends
6. Complementary sticky-ended fragments are annealed by hydrogen bonds
7. DNA ligase is added to join the fragments and plasmid together
8. With the formation of phosphodiester bonds

b) Account for the results as shown on the replica plates. [3]

1. Colonies 1, 2, 3 and 6 are made up of bacterial cells that have taken up a plasmid
2. They have the intact ampicillin resistance gene and will be resistant to ampicillin / can grow in the presence of ampicillin
3. Colonies 1 and 3 are made up of bacterial cells with intact tetracycline resistance gene
4. They are resistant to tetracycline and hence they have taken up the reannealed plasmids
5. Only colonies 2 and 6 have taken up recombinant plasmids with successfully integrated bovine somatotrophin gene
6. As the gene inserted disrupted the tetracycline resistance gene

c) Discuss which method should be used by the company to increase milk production in cows. [4]

1. Method 1 / Injection of hormone into cows
2. Although the volume of milk produced by 2 years old cows with injected hormone is less than that of 2 years old genetically modified cows by 4.6 million gallons in a year
3. The method is less invasive
4. and easier to carry out
5. As adult stem cells are difficult to find
6. And culture in lab
7. There is less ethical and social issues as compared to using genetically modified cows

OR

8. Method 2 / Genetically modified cows
9. 2 years old genetically modified cows produced 4.6 million gallons more milk than cows with injected hormone
10. And it results in long term production of milk
11. Thus there is no need to inject hormones at intervals
12. But the method is more invasive
13. and harder to carry out
14. As adult stem cells are difficult to find
15. And culture in lab
16. There is more ethical and social issues as compared to injecting rBST into cows

[Total: 10 marks]

SECTION B: FREE RESPONSE QUESTION

QUESTION 5

(a) Explain how the structure of haemoglobin and glycogen are related to their function. [8]

HAEMOGLOBIN	
Structure	**Function**
S1. Formation of a hydrophobic cleft, containing a haem binding site	F1. This provides a hydrophobic environment for the haem group to function
S2. Each subunit bears 1 haem prosthetic group/ each haem group bears an Fe^{2+} ion within a porphyrin ring	F2. Fe^{2+} ion is able to bind reversibly to oxygen accounting for the oxygen-transporting ability of haemoglobin
S3. Amino acid residues found on the surface are generally hydrophilic	F3a. This allows haemoglobin to be a soluble globular protein in aqueous medium
S4. Binding of oxygen to 1 of the 4 subunits causes the haem group to change conformation, resulting in further conformational changes in the remaining subunits	F4a. This allows the other subunits to more readily bind to oxygen

GLYCOGEN	
Structure	**Function**
S1. It is a large molecule	F1a. It is Insoluble
S2. It is composed of several hundreds to thousands of glucose monomers	F2. The large number of glucose molecules act as a large store of carbon and energy
S3. Glucose units linked by α(1,4) glycosidic bonds forming helical coils	F3a. α(1,4) glycosidic bonds result in helical coil so that structure is more compact and ideal for storage
S4. Glycogen is highly branched due to the presence of α(1,6) glycosidic bonds	F4a. The compact shape allows for easy storage
S5. Anomeric carbon involved in glycosidic bond formation leaving few free anomeric hydroxyl groups	F5. This makes glycogen an unreactive and chemically stable compound, ideal as storage substance

b) Outline the mode of action of restriction enzymes. [6]

1. Mode of action of restriction enzymes is based on induced fit hypothesis
2. Active site of restriction enzyme recognise and bind to restriction site on DNA
3. Resulting in formation of restriction enzyme-DNA complex
4. The binding of DNA induces a change in 3D conformation of the active site
5. DNA is brought in close proximity and correct orientation / fit the substrate snugly into active site of enzyme
6. to the catalytic amino acid residues in the active site of restriction enzyme
7. This results in the stretching of critical bonds of DNA
8. and stabilization of the transition state
9. and lowering of activation energy of the reaction
10. Restriction enzyme will cleave the double stranded DNA
11. By breaking phosphodiester bonds
12. Resulting in formation of blunt or sticky ends

c) Discuss the effect of inhibitors on the rate of enzyme activity. [6]

<u>Competitive inhibitors</u>

1. Competitive inhibitors are structurally similar to the substrate molecule
2. and compete with the substrate for binding to the active site
3. Competitive inhibitors remain bonded to the active site
4. And prevents substrate binding to active site
5. There is less enzyme-substrate complex formed per unit time
6. Therefore, the initial rate of reaction increases more slowly in the presence of competitive inhibitor
7. V_{max} of an enzyme is reached at a higher substrate concentration in the presence of competitive inhibitors

<u>Non-competitive inhibitors</u>

8. Non-competitive inhibitors bind to a site other than the active site of enzyme
9. and induced a change in the overall 3D conformation of the enzyme
10. and the 3D conformation of the active site
11. Conformation of active site is no longer complementary to the substrate
12. and substrates cannot bind to active site of enzyme
13. The initial rate of reaction increases more slowly in the presence of non-competitive inhibitor
14. Even when substrate concentration is very high, the initial rate of reaction does not reach the same V_{max} as in the absence of non-competitive inhibitors

QUESTION 6

(a) Describe the normal functions of embryonic stem cells and blood stem cells in a living organism. [8]

Specific function of embryonic stem cells

E1. Found in the inner cell mass of the blastocyst
E2. Embryonic stem cells are pluripotent
E3. are capable of developing into all fetal tissues/f orm all the cells of the embryo
E4. give rise to three primary germ layers/ ectoderm, mesoderm, endoderm
E5. which can differentiate into specific cell types via differential gene expression

Specific function of blood stem cells

B1. Hematopoietic stem cells are found in the bone marrow
B2. capable of long-term self-renewal through mitosis
B3. replenishing their own population
B4. are multipotent
B5. can differentiate to give rise to all the blood cells/red blood cells and white blood cells in the adult human
B6. function in maintenance and repair/replace blood cells that have die because of injury or disease
B7. AVP/Any valid example

(b) Explain how meiosis and random fertilization can lead to genetic variation. [6]

Crossing over during prophase I

1. formation of chiasmata allows exchange of genetic material between homologous sections of non-sister chromatids
2. results in recombination of segments of non-sister chromatids/recombinant chromatids OR giving rise to new combinations of paternal and maternal alleles

Independent assortment during metaphase I

3. in metaphase I, pairs of homologous chromosomes align randomly at the metaphase plate
4. gametes formed have 50% chance of receiving paternal or maternal chromosome/alleles
5. which will result in new combinations of chromosomes/alleles
6. number of possible combinations of gametes due to independent assortment is 2^n; for a diploid organism, where n = number of sets of homologous chromosomes

Random fertilization

7. As gamete from each parent can carry a wide variety of different combinations of alleles
8. fusion of two gametes will result in a zygote / an offspring with new genetic variation

(c) Explain with a named example, how environmental factors act as forces of natural selection. [6]

1. A named example, e.g., peppered moth / industrial melanism OR handedness of (scale-eating) Cichlid fish OR Darwin's finches
2. Reference to existing variation in isolated populations
3. Reference to environmental factors exerts selection pressure
4. Reference to fitness to the environment are selected
5. Reference to the outcome of natural selection
6. Individuals fitted to the environment survive to reproductive age, so that favourable alleles are passed down to offspring, and over time sufficient changes in allele frequencies occurs